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yeah all good now okay yeah somebody said devops first principles that is our entire um i would say cohort sprint right what is the today's topic networking and docker yes so can i depend on irfan again is irfan available at least yeah i can see irfan okay initially i will share the screen let me start my camera i hope i'm visible as well okay what is the network what is networking or what is the network let's start with some question first yes what is the network connection of nodes yes uh path for communication yes there is also a right answer why do we need networking by the way this version is also right answer sharing information between them why do we need networking to transfer data we can transfer data by copying into usb and we can share right sharing data it's connectivity but yes but why do you need connectivity yes when is the data but the most more important is you can you can share data uh between devices this is i would say more precise because you can share data on the same machine as well right so between two devices and other thing is uh a shared compute right the shared compute shared memory or shared uh like data as well so not just not just data we can have a shared computer that's the reason we have distributed system as well right so we can offload some computing some processing to other system yes so before networking we had inter-process communication and we used to communicate with the uh basically two types of uh two ways of communicating with the other applications i would not say other devices but other applications so what is the shared memory how many of you know shared memory so those who have worked on in in c language sh memgate uh is the function that we used to use hh memgate right so basically your os used to create a specific memory area so that two process would uh going to you know access that area for share the data but then it was very very tindersome very very very tedious very clumsy and then the next invention specifically this is from bsd freebsd which is berkeley software distribution of linux that's where your sockets are implemented right the first sockets given by given to the world is by bsd linux machine and that's where the concept of pipes uh across the network so initially it was pipes only based on the same devices like we call it the ipc inter-process communication by using pipes but then the same technology extended with the uh pipes using the okay but when i say a network in the technical world for example if i'm saying i i'm going to let's let's assume i'm doing subnetting how many of you have at least heard this word subnetting how many of you know subnetting or hard disorders yes subject mask i'm talking about yes you're right right so if i if i say that i am going to uh you know do some network connectivity in my application it could be rest api could be something and there is a challenge there is some issue there is some bug whatever you call defect how you would uh start you know uh diagnosing about it what is the basic information you know again i i don't want to like uh answer like wire sharp or something i mean those are the tools i can understand or maybe if you are working on web development you're working on uh you might know developer tools right but uh let's say you want to diagnose network packet you want to diagnose network packet what is the bare minimum information you must know so that you can start working on networking or any networking related

challenges so if i i ask you show me your network in your system yes dns ip address what else yes that's a perfect that's one one part of the answer what else yes so dns ip address port yes great what else mac address yes yes what else gateway someone said yesterday communication gateway right so gateway a domain is anyway dns uh part of domain so port mac ip address and gateway and submitting a mask so these are the core networking topics i would say that one should understand okay now uh one more question i just wanted to grab the you know a high level of audience so that i can i will start accordingly what should i need to cover what is the port number for http or http application in general what is the port number http is 80 yes how many of you like okay only venkatesh and rajat knows this what about others okay for http yes yes so hdb is 80 and https is 443 if i ask why why the port number for http is 80 and hdbs is 4443 what would be your answer smdb is 25 yeah but why that standard standard comes for some reason yes you are right going conditional from where this standard or chronology comes into the picture where is a so if we consider osi model let osi model for like seven layers where this port number comes into the picture look at the guys again we have seen this uh since day one right when i have i used to ask question why we are circumventing around it that htp is old 80 were already occupied yes i got to know that they are standard code their chronology the question is specifically around why why anyone has decided that port number should be 80 port number should be 4443 if there's smd gateway smdb gateway then port number 25 i guess smdb gateway ssh port number 22 right the question is why yes uh but then i can also say any port right so all the devices know that http communication happen on particular port why just 80 default and that application layer uh answer is again incorrect okay so port number is on transport layer port number is on transport layer so basically your http is actually underlying connection is either tcp or udp right so on http when you say http there is a two way you can uh make the connection to your server one is tcp or one is udp okay so great so i i got a bit of idea uh like uh what is overall understanding so let's get started with a very high level basics of osi model so that if you face any issues as such you must be knowing a very a broader level of picture of osi okay because application layer is the at top so it start with physical it's it's the last layer so physical layer is the first layer then it's a data link third third layer is networking fourth is transport fifth is session sixth is presentation and seventh is application right now this uh session presentation and application are like merged together and that's the reason we call it as a one application layer yeah you're right so if you are saying opposite yes uh so from the context context of stack so all these layer are stacks on top of each other so in that case application layer would be a last in first out so it would be a okay great are we good so far uh let's let's start with some uh basic knowledge on the osi and then we'll start with the docker networking okay great again and for this as well we need to go to very very detailed level of basics so let me ask the question uh what is what is the baud rate you know baud rate b-a-u-d baud rate bits per second yes okay so good we have good audience actually how many bits makes one bytes great great why why one byte equals equal to eight bits because from from that point of time we need to pick up you to understand the first layer of osi okay two two raised to the power three i can say eight but why only eight then why not 10 why not 16 i don't know have you never been like curious about why why such things has been like this way

only because yes yes yeah something to do with physical structure but why eight machines are by the region it can be a two it can be four eight sixteen why only eight yes yes but they they created of eight bytes that's the reason i'm asking why memory of race is filled instead of eight bytes from where this head is coming why did you created from 16 by 16 bytes or 32 bytes okay so let's start from that point okay because that's the core of your physical world okay and then based on this idea uh your physical world has been evolved okay so we'll start with the story imagine imagine uh you all are scientists okay i know it's very hard to imagine but we need to imagine okay you are all our scientists and i am your assistant okay i am your assistant and uh since you you have a lot so generally scientists are pretty busy on research and something right so they need someone as an assistant and you want first to check assistant is really good or not like they would like to check the knowledge so you started asking me what is 8 plus 8 right uh tell me the table of 10 and i mean that much maths i know so i can i can probably say that i can give you the right answer right so initially you want me to uh do some simple analytics and once you trust me that's where you would like to give me let's linear equation and so on and so forth right so what is the moral of this story let me it's better to you know i can draw and then we can go in more bits and pieces of networking let me share my screen by the way i have operated docker uh entire material uh into our repository let me start with not new sorry my bad it was which one was this the one yes all right so i was saying that you are going you have to give me let's say if you ask me addition right so if you want me to do addition what bare minimum requirements you need to provide me if you want me that i should be able to do the addition what you will give to me you would you would at least need to provide me a numbers two numbers at least right inputs so uh how many numbers we have 0 to 9 yes we have 0 to 9 numbers so let's say 0 to 9 means 10 plus 0 to 9 is 10 right what else uh today you will give me an addition but tomorrow you might ask me subtraction division modulo counting uh basically uh multiplication right so then that is what we call as symbols so plus minus division yes tomorrow you also would like to trust me on currency level conversion so we have in total we have 32 symbols plus minus addition subtraction all counting all right square square root and all those things what else if you ask me to do uh linear equations uh let assume x equal to something or then y equal to nx plus c right so in that case you would i would require knowledge of alphabets both capital and small so 26 capital 26 small yes are we are we good guys so far yes so total the total information is so total thing i mean the way you want me to properly give you all this information so if i make the total of this i guess is close to something around 94 if i'm not wrong right so you need to pass me the 94 as a piece of information yes and since this minimum piece of information you need to pass me how i can factor in that information with the help of binary format so if i consider uh the binary why binary because machine understand machine understand binary by the way machine understand power language so voltage on voltage off yes that's the only language machine understand right and that's the reason power on represents one and power off represents zero and that's the reason so it's understand binary okay so binary our representation machine doesn't understand binary just to make it super clear it understand power and power off okay so if i need to if i now understand that i need to uh at least provide our research assistant mohit being we are a scientist i need to

provide 94 bare minimum piece of information then i need to calculate 2 raised to in 2 raised to 1 right so 2 4 8 2 4 8 16 32 64 and 127 right so 120 sorry 128 so 128 is greater than 94 are you still with me guys 52 are alphabets 26 capital 26 small okay okay so you must pass me at least information of the at least 94 some of this all of this right i would digit i would need uh symbols like plus minus modulo division that's what bare minimum requirement right that if you ask me if you ask me so this 8-bit has some history right i will come to that why 94 because 56 52 are alphabets 26 small 26 capital 32 symbols plus minus and dollar and what next right and 10 are digits 0 to 9 so i need to fit this entire information so how many bits will help me total bits so that i can get work delivered from mohit yes so mohit is not a normal person okay he he only understand this power on power off yes so then if i go into the uh index of 2 so 2 4 8 16 32 64 128 right so 7 piece of information yes so if i say 2 raised to power 7 i will get 128 yes with me so far so ideally 7 bits uh 7 bits were sufficient for a byte but then let's make a i don't know guys come on this is really a first year of engineering okay i don't know why i need to spend so much time but no worries i'm okay i mean because i need that kind of foundation super clear to let you understand why it is 8 bit but then uh let's okay let's do one thing now you can see me but uh if you put me in some different room room okay and uh let's we have this binary language so you in my room there are 8 lamps or 7 lamps okay and you are passing information by turning on those buttons and so first you will on let's say 20 and uh next time you let's say on 25 but this time i didn't answer uh addition as 45 i have answered something else something weird let's say 55 or 50 whatever but my answer was not right and now that creates uh some doubt in your mind some skepticism in your mind that is this gonna work properly and then uh maybe it's a case that uh that while you are entering 20 plus 25 i might receiving 20 plus 30 why because they're they're kind of there may be an electrical circuit shock or short right and that's the reason uh to let us know like what if there is an electrical short circuit or shortage of something how we know that what information you pass to me is really exactly same information i am receiving and for that one more bit is added called if anyone from you are electronics engineer the the bit is called parity bit yeah it's kind of checksum but that bit is called parity bit and that makes your all memory chips in the range of 8 bits so at that time they have added this and that makes this as 8 bit but later this 8 bit has been uh like this error bit like a parity bit has been shifted to a 3 bit cell called ecc error correction code but since the chip were of 8 bit it was anyway uh in the in the it's working like same thing right so this is how uh you have a memory i hope right now you understand why this 8 bit okay hopefully okay great so then why we are discussing about this we'll come to that part now so ultimately you need to share the data between the computers yes that's what networking is all about right you need to share the data information and as well the computation because sometimes server server is running you are processing some request and then you get the response at the client side right so data is for sure there but then there is compute as well and even that is the basic foundations of your cloud computing that distributed compute so uh i think we have already seen this uh why okay now okay let's i think we have discussed this when i say a equal to 97 and a equal to 65 why is that let me answer you your answer let me answer this first okay i need some different answer a equal to 97 b equal to 98 why is that we are unknowingly

started working on physical layer okay yeah that that i understand that difference between capital a and small a my question is about the number why a equal to 65 uh capital y equal to 97 from why ascii has decided like this what was the reason okay let me simplify this question any anyone have a mechanical keyboard it's it it is like an very annoys with its sound if someone from gaming background they used to have a mechanical keyboard and uh i personally like the sound of the mechanical keyboard but a lot of them my wife gets very irritated and i know it so uh why why i ask medical keyboard so every uh key you know every key of the keyboard underneath it you have you have you have some circuit so if you let's if i ask you to take z z out and a out the key z and a i'll take it out and swap okay so now a becomes z and z becomes a okay and i ask you type in notepad so what do you think when i type z it will still type z no why no so so z z becomes a right even with that z becomes a and a becomes z so the reason the reason the reason is the reason is because excited to understand this why it is like that okay so listen this imagine okay always reverse it if i don't put it like this okay a equal to 97 b equal to 98 so one assumption is good that uh they have to send that value right device only understand binary binary so need they need to map on something but then this is this is again answer of what not why okay so we need to understand you can only understand why when you don't know

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